

NIZHNEVARTOVSK, Russian Federation

WMO: 234710

Lat: **60.949N**Lon: **76.484E**Elev: **177**StdP: **14.60**Time Zone: **5.00 (E05)**Period: **04-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -40.1	(c) -34.7	(d) -45.4	(e) 0.4	(f) -38.9	(g) -39.8	(h) 0.6	(i) -35.0	(j) 20.0	(k) 10.9	(l) 18.7	(m) 5.0	(n) 5.1	(o) 40	(p) 0.723

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.7	83.5	66.6	80.3	65.6	76.8	64.2	69.6	79.0	67.6	76.1	65.6	73.9	9.0	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 66.4	(b) 97.8	(c) 73.7	(d) 64.4	(e) 91.2	(f) 72.0	(g) 62.5	(h) 85.1	(i) 70.2	(j) 33.8	(k) 79.3	(l) 32.1	(m) 76.7	(n) 30.6	(o) 74.2	(p) 77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.0	18.2	16.5	-42.0	88.3	7.3	2.9	-47.2	90.4	-51.5	92.1	-55.6	93.7	-60.9	95.9
			-41.7	73.1	6.5	2.2	-46.4	74.6	-50.2	75.9	-53.9	77.2	-58.6	78.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.9	-6.5	-1.7	17.7	31.8	42.6	59.8	64.2	58.0	47.2	32.0	11.6	0.4		
	DBStd	26.96	17.28	15.26	11.55	10.80	9.87	9.37	6.97	6.75	7.82	10.11	15.53	16.74		
	HDD50	8453	1750	1447	1000	549	279	23	1	8	146	560	1152	1538		
	HDD65	12949	2215	1867	1465	996	698	208	102	235	535	1023	1602	2003		
	CDD50	1129	0	0	0	3	48	316	442	255	62	3	0	0		
	CDD65	149	0	0	0	0	2	51	77	18	1	0	0	0		
	CDH74	1297	0	0	0	0	41	487	648	117	4	0	0	0		
	CDH80	282	0	0	0	0	7	115	149	11	0	0	0	0		
	WSAvg	8.2	7.6	7.5	9.3	9.2	9.1	8.5	7.3	7.1	7.8	8.9	8.6	8.1		
	PrecAvg	31.2	1.9	1.6	2.1	2.0	2.6	3.0	2.8	3.3	2.7	3.4	3.2	2.6		
(15) Precipitation	PrecMax	39.5	3.4	2.9	4.8	3.2	4.6	5.9	5.7	5.0	4.5	5.6	5.0	4.6		
	PrecMin	25.4	0.5	0.5	0.3	0.2	1.3	1.1	1.3	1.6	0.9	1.5	1.5	1.1		
	PrecStd	3.9	0.7	0.6	0.9	0.7	0.9	1.0	1.0	1.1	0.8	1.0	0.9	0.9		
	PrecStd	3.9	0.7	0.6	0.9	0.7	0.9	1.0	1.0	1.1	0.8	1.0	0.9	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.7	34.1	42.9	62.9	79.1	87.5	87.9	81.0	73.5	59.4	39.6	32.2		
		MCWB	31.4	33.2	36.1	51.4	58.8	66.9	69.8	68.9	59.9	51.1	38.9	31.7		
	2%	DB	26.2	28.7	37.8	53.9	72.0	82.8	84.2	77.0	68.0	52.0	35.9	30.0		
		MCWB	25.6	28.0	33.0	45.3	54.5	65.3	67.2	64.8	58.2	46.6	35.4	29.3		
	5%	DB	21.4	21.7	35.5	49.8	65.9	80.4	80.7	73.6	62.9	48.1	33.8	26.7		
		MCWB	21.0	20.4	32.1	42.7	53.2	64.5	66.7	63.4	55.9	45.2	33.4	26.2		
	10%	DB	17.7	18.0	33.3	46.0	59.2	75.8	77.3	69.9	59.1	44.7	31.9	22.9		
		MCWB	17.0	16.8	30.8	40.1	49.3	63.0	65.4	62.1	53.4	42.7	31.4	22.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.9	33.3	36.8	51.5	60.3	70.6	73.1	70.9	62.6	51.6	39.0	31.9		
		MCDB	31.1	34.1	41.3	60.9	74.3	80.9	82.7	79.2	70.0	57.7	39.3	32.0		
	2%	WB	25.1	28.2	34.0	46.4	56.7	68.1	70.5	67.5	59.3	48.2	35.4	28.8		
		MCDB	25.5	28.8	36.4	53.0	69.2	79.0	80.0	73.6	65.3	50.9	35.8	29.3		
	5%	WB	20.6	20.8	32.4	43.1	53.9	66.0	68.6	65.1	56.9	45.1	33.3	26.1		
		MCDB	21.1	21.8	34.5	48.4	64.0	76.6	77.3	71.1	62.2	47.2	33.6	26.6		
	10%	WB	16.8	17.0	30.4	40.2	50.4	64.1	66.9	62.9	54.4	42.6	31.1	22.1		
		MCDB	17.5	18.0	32.9	44.8	59.2	73.8	74.9	68.6	58.6	44.4	31.5	22.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.6	14.5	16.6	16.2	16.9	17.7	17.7	15.8	13.3	9.8	11.9	13.3		
		MCDBR	12.8	13.5	15.2	19.9	25.7	23.4	22.7	20.9	18.7	13.8	8.9	13.1		
		MCWBR	12.9	12.8	12.2	13.3	13.9	11.4	11.1	11.5	11.0	9.7	8.9	13.0		
	5% WB	MCDBR	13.0	13.6	14.4	18.5	24.2	20.5	19.9	19.1	17.3	12.9	8.8	13.2		
		MCWBR	13.1	13.1	12.3	13.2	14.5	11.1	10.7	11.7	10.9	9.7	9.0	13.2		
(40) Clear-Sky Solar Irradiance	taub		0.259	0.302	0.354	0.374	0.387	0.402	0.437	0.410	0.361	0.329	0.268	0.245		
	taud		1.966	2.005	1.952	2.069	2.284	2.332	2.210	2.270	2.360	2.238	2.018	1.907		
	Ebn at Noon		137	203	231	256	265	262	247	244	237	197	131	85		
	Edh at Noon		14	26	39	42	37	36	40	35	26	21	13	10		
(44) All-Sky Solar Radiation	RadAvg		126	410	895	1405	1538	1724	1623	1127	709	306	124	63		
	RadStd		10	35	56	138	128	142	186	137	87	36	27	6		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page